



Research paper

Theoretical insights into the structural, electronic and gas-sensing properties of Sc-decorated black phosphorus via strain engineering

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A B S T R A C T

Structural and electronic properties of Sc-decorated black phosphorus (SP) and its gas sensing for NO molecule were investigated using density functional theory with the emphasis on the response to applied strain. It found that SP transformed into indirect bandgap semiconductor, and converted into a metal characteristic under strain. SP showed strong interaction with the NO with a high adsorption energy of -2.92 eV since the orbital hybridization and extensive charge transfer, and exhibited 3.3-fold sensitivity compared to black phosphorus, suggesting that SP was a potential NO sensor. Furthermore, strain engineering provided an effective route to facilitate the desorption of NO.

1. Introduction

Two-dimensional black phosphorus (BP) has received widespread attention due to the excellent electronic and sensing properties, such as ultra-high carrier mobility, tunable band structure, high surface-to-volume ratio and remarkable sensitivity towards gas molecules, making it promising for applications in gas sensors and other field [1,2]. Particularly, BP was found to be sensitive towards the nitrogen-based molecules [3]. Despite some pioneering efforts, the development of gas sensors based on BP was still in its emerging stages. One of the main challenges lied in the inherent instability of BP, which could significantly hinder the sensor performance. To improve the stability of BP-based sensors and avoid degradation under normal atmosphere, many researchers have proposed strategies such as non-covalent functionalization. It allowed BP to showcase its inherent properties without the need for operating sensors in inert or vacuum environments [4]. Another challenge was that BP exhibited weak capture ability of nitrogen-based gas molecules due to the low adsorption energy [5,6]. Hence, it was necessary to enhance the stability and sensing performance towards gas molecule by using some effective methods [7–9].

Atomic decoration/doping theoretically and experimentally have been demonstrated to be an important way for modulating the properties of 2D materials [10–12]. For instance, in Bafekry et al.'s work, BC₂ doped with N, As and P atoms could dramatically alter the electronic structure, bolster the dynamic stability of different 2D materials [13] Caporali et al. experimentally demonstrated the ambient stability and

functionality improvement of nickel-decorated BP films [14]. Among them, scandium (Sc), as the simplest rare earth and transition metals, have been used as dopants in the semiconductor materials. Some studies indicated that Sc-doped BP demonstrated superior stability for a substantial decrease in the conduction-band minimum (lower than the redox potential of O₂/O²⁻), and a strong chemical interaction between Sc and BP [15,16]. The results of another research showed that Sc-doping improved the adsorption effect of Sulfur-based gas on the MoS₂ surface [17]. Particularly, a report by Azam Marjani et al. has shown that intrinsic BP under Sc doping could be transformed into a highly sensitive hydrogen sulfide detector and exhibited improved functional sensitivity [18]. Thus, it could be inferred that the introduction of Sc had a high probability to be a good sensing material for harmful gases detection.

As we all know, strain engineering, as an inevitable factor in material applications, held great significant importance in altering the physical properties of two-dimensional materials [19]. It has been shown that BP could withstand up to 30 % strain [20]. Additionally, the electronic and optical properties of BP were highly sensitive to the magnitude and direction of applied strain [21]. It was discovered by Cai et al. that strain altered the adsorption energy, magnetic moment and magnetic coupling state of transition metal atoms adsorbing on BP [22]. Another study has demonstrated compressive strain could effectively regulate the amount of NO₂ adsorbed on BP sensor [23]. Yang et al. found that applying strain to transition metal-doped BP had a significant impact on the adsorption behavior of nitrogen oxide (NO) molecule [24]. Inspired by

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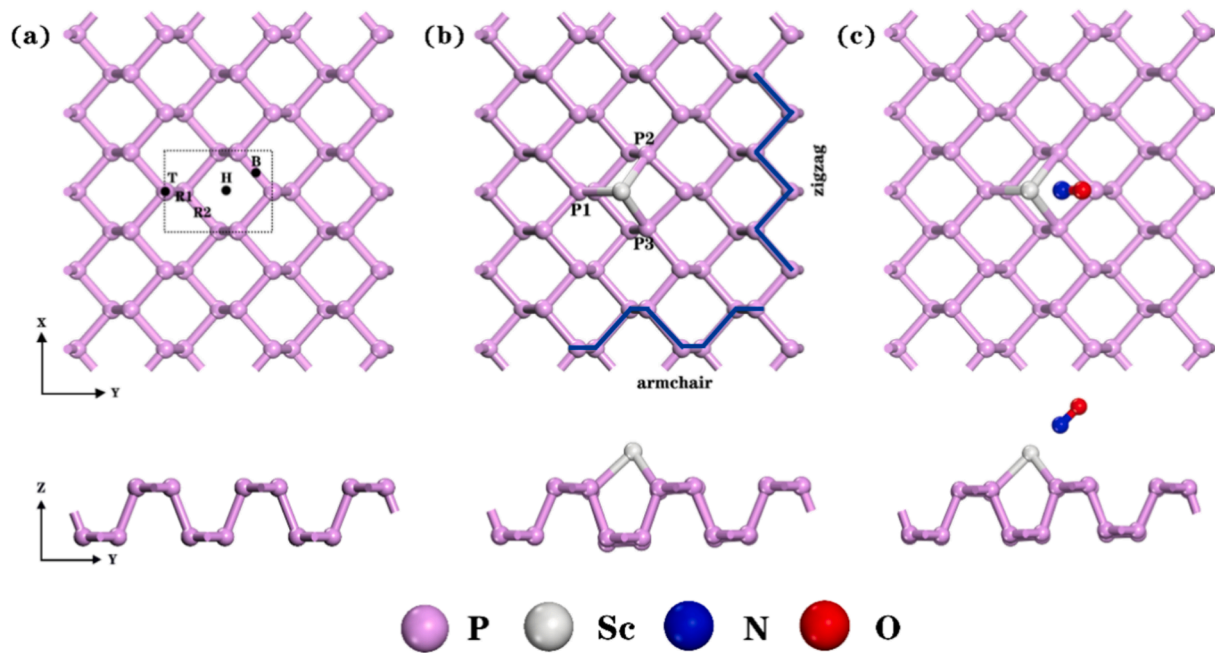


Fig. 1. (a) Optimized geometrical structure of pristine BP and three typical positions of H, B, T. Geometrically optimized structure of (b) Sc decorating pristine BP and (c) NO gas adsorbing SP.

these findings, it was anticipated that strain could control the electronic properties and induced tunable response in gas sensing performance on the Sc-decorated BP.

Many previous studies have not involved the relevant research of the modulation mechanism of strain engineering on the electronic properties and gas sensing of Sc-decorated BP. Therefore, this paper aimed to study the influence of biaxial strain on the electronic properties of Sc-decorated BP and its sensing ability towards NO gas molecule. Results revealed that a strong strain dependence of the geometric structure and electronic properties of Sc-decorated BP. It demonstrated a significant enhancement in adsorption strength and sensitivity for NO gas molecule. Furthermore, biaxial strain engineering controlled the adsorption and desorption of NO gas molecule on Sc-decorated BP. This work provided a theoretical foundation for Sc-tunable spintronic devices and novel gas sensors with enhanced operability.

2. Computational details

This study was based on the density functional theory for all the structure relaxation and electronic properties calculations, which were implemented in the Cambridge Sequential Total Energy Package (CASTEP) code [25]. The exchange–correlation energy and core electron were described by the Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional in the generalized gradient approximation (GGA), and previous studies have demonstrated the accuracy of GGA in predicting the magnetic states of transition metals [26,27]. Hubbard U value for the d-orbital of the Sc atom was set to 2.5 eV to precisely depict the electronic properties of the systems [28]. The spin-polarization of atomic configurations was included. The van der Waals (vdW) interaction was considered with dispersion correction via the semi-empirical correction scheme of the DFT-D2 approach, with better prediction of adsorbing energy and distance [29]. There were also extensive applications of DFT-D2 in research of gas adsorption [30,31]. To eliminate the interaction caused by the periodic structure, a $4 \times 3 \times 1$ supercell containing 48 atoms was selected, with a 20 Å vacuum layer perpendicular to the substrate. Additionally, the Monkhorst-Pack network method was used to sample the Brillouin k-point in the reciprocal space, and the network size of structure relaxation and electronic properties

computations were set as $3 \times 3 \times 1$ and $5 \times 5 \times 1$, respectively. Ultrasoft pseudopotential was used, and the energy cut-off was set to 400 eV. The energy, force, stress, and displacement tolerances were lower than 10^{-5} eV/atom, 0.01 eV/Å, 0.01 GPa, and 0.003 Å, respectively.

The applied strain was simulated by adjusting the lattice constants of the supercell in the armchair and zigzag directions [21]. The strain value was defined as the relative change in lattice constants:

$$\varepsilon = (\alpha - \alpha_0) / \alpha_0 \quad (1)$$

where ε represented the magnitude of strain deformation, α_0 and α referred to the strained-free and strained lattice constant, respectively. Accordingly, negative values were attributed to compressive strain, while positive values were assigned to tensile strain.

The structural stability of Sc-decorated BP could be analyzed using a binding energy (E_b) formula [32], which could be determined by the following equation:

$$E_b = E_{Sc+BP} - E_{Sc} - E_{BP} \quad (2)$$

where E_{BP} , E_{Sc} , and E_{Sc+BP} were the total energy of the BP, the energy of isolated Sc atom, and the energy of the Sc-decorated BP, respectively. Negative E_b donated the stable decoration system.

In order to determine the adsorption capability of different systems for the gas, the adsorption energy (E_{ads}) was defined as follows [33]:

$$E_{ads} = E_{total} - (E_{sub} + E_{NO}) \quad (3)$$

where, E_{total} represented the total energy of the adsorption system, while E_{sub} and E_{NO} represented the energy of the pristine/Sc-decorated BP substrate and the energy of NO gas molecule, respectively. A stronger interaction with gas was demonstrated by the significant value of negative E_{ads} .

3. Result and discussion

3.1. Structural and electronic properties of Sc-decorated black phosphorus

In Fig. 1(a), the optimized geometry of pristine BP was displayed. The lattice constants of the relaxed BP were $a = 3.37$ Å and $b = 4.59$ Å,

Table 1

Mulliken population analysis, bond lengths, and distance between N atom and the nearest atom, as well as adsorption energy/binding energy (E_b/E_{ads}) for different systems.

System	E_b/E_{ads} (eV)	Bond	Distance (Å)	Overlap population
SP	−3.41	P1-Sc	2.49	0.48
		P2-Sc	2.55	0.39
		P3-Sc	2.55	0.39
BP-NO	−0.21	P1-N	2.35	0.05
SP-NO	−2.92	N-Sc	1.92	0.53
		P1-Sc	2.61	0.37
		P2-Sc	2.62	0.37
		P3-Sc	2.62	0.30

and the P-P bond lengths were $R1 = 2.26$ Å and $R2 = 2.24$ Å in the vertical and horizontal directions, respectively, consistent with the literature description [34,35]. As shown in Fig. 1(a), three typical decoration sites of H, T and B were considered. Nonetheless, Sc adsorption on B and T sites were not stable after optimizing, potentially due to the introduction of Sc disrupting the folded hexagonal structure

of the P-P bond, ultimately leading to structural collapse. It was consistent with results from other studies that H site had larger binding energy and more stable geometric structure [36]. Therefore, H site was selected as Sc-decorated position for subsequent research. Additionally, different initial adsorption distances (2.5 Å, 3 Å, 3.5 Å) were tested to mitigate the occurrence of local minimum. Ultimately, the intermediate value of 3 Å was adopted between substrate and adsorbate.

The optimized adsorption structure was shown in Fig. 1(b) with 2.08 % coverage. In the Sc-decorated black phosphorus (SP), the P2-Sc and P3-Sc bond lengths were equal and slightly longer than the P1-Sc bond as listed in the Table 1 due to the folded structure of BP. Importantly, the structure still maintained integrality, illustrating that the inherent excellent properties of BP could be preserved. The adsorption of Sc on BP had closer distance and a larger binding energy (−3.41 eV) than other transition metal atoms [37], demonstrating the better adsorbing stability of the SP structure. It was found that theoretical cohesive energy of bulk Sc was calculated to be −3.31 eV [38]. The absolute value was lesser than the theoretical binding energy of SP, which suggested that Sc atoms may not exhibit aggregation. To further demonstrate the dynamic

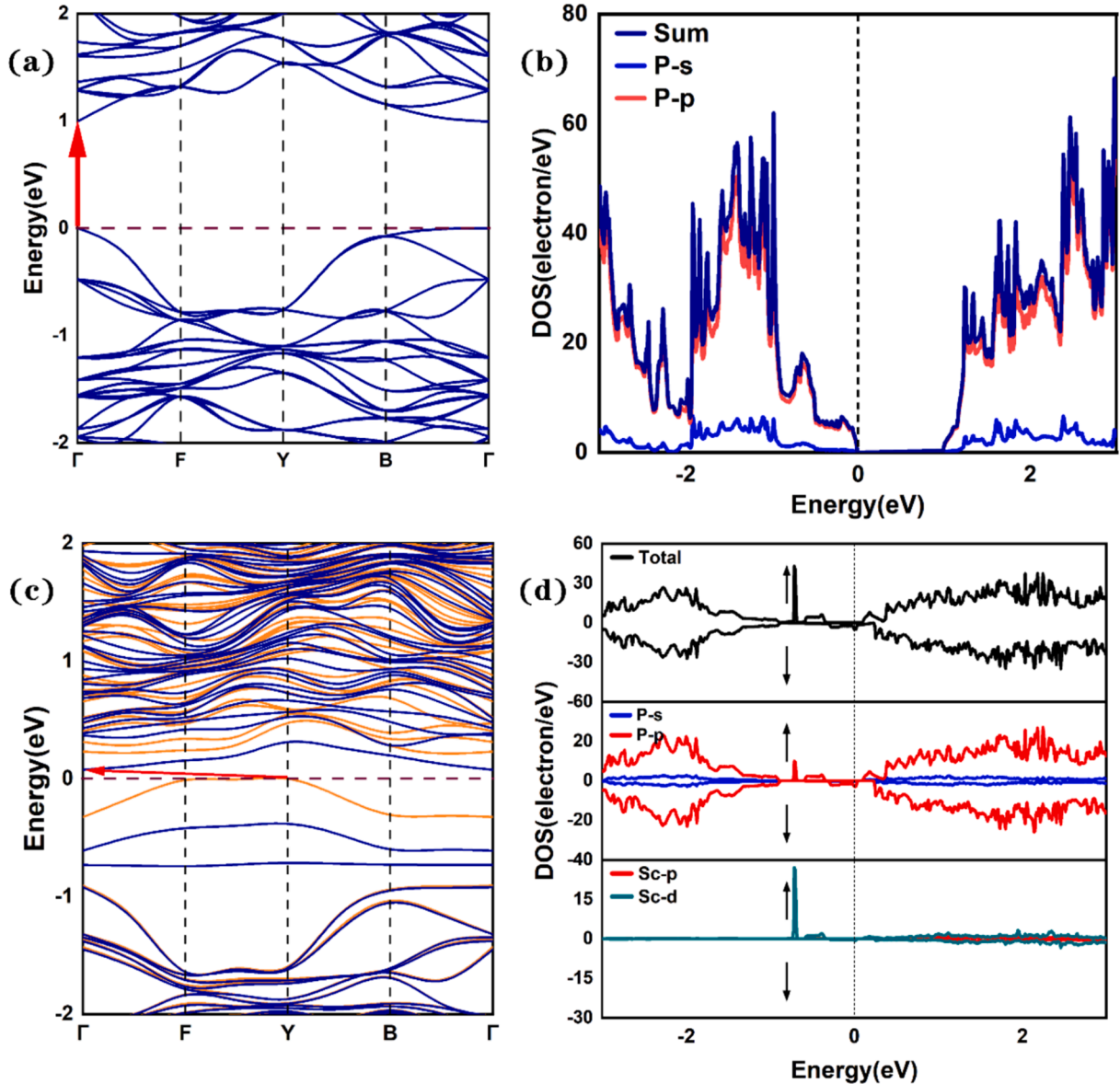


Fig. 2. (a) The band structure and (b) DOS of pristine BP. (c) The band structure and (d) spin polarized DOS of SP. The dash line in the DOS illustrated the Fermi energy. Blue represented the up-spin band, while orange represented the down-spin band in the band structure.

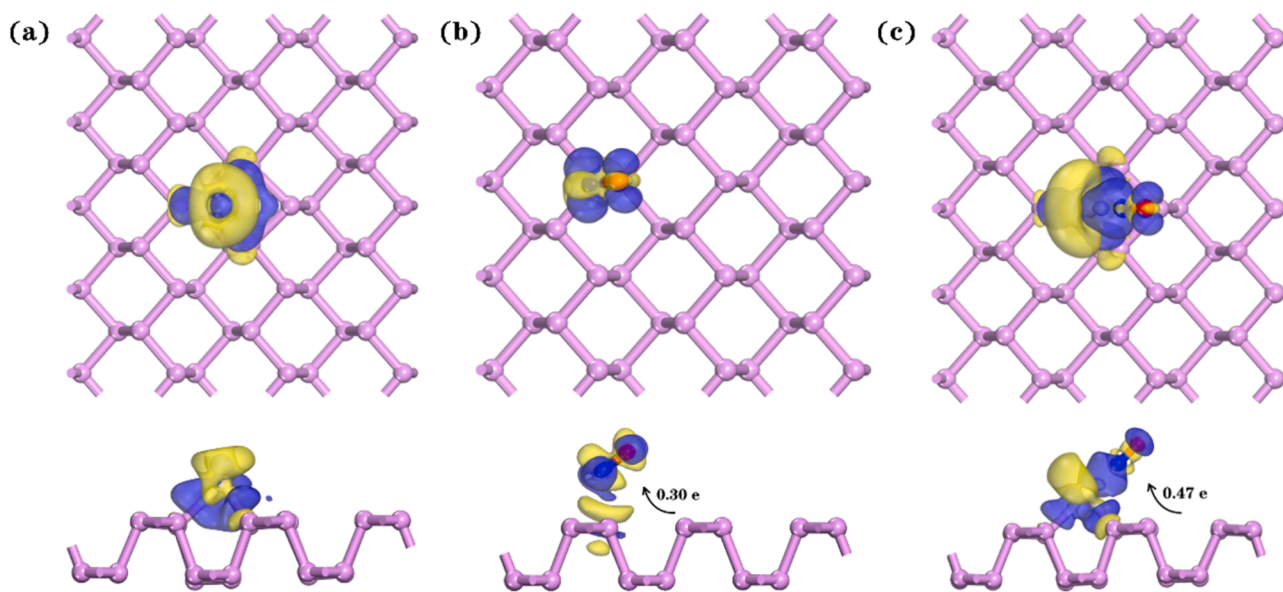


Fig. 3. The electronic density difference of pristine BP adsorbing (a) Sc atom, (b) NO molecule and (c) SP adsorbing NO molecule. Blue and yellow represented an accumulation and loss of charge, respectively. The isosurface value was taken as $0.02e/\text{\AA}^3$.

Table 2

Under biaxial strains ranging from -8% to 8% , the P-P bond lengths of BP in both vertical and horizontal directions (R1, R2), the binding energy, the magnetic moment and band gap of SP.

Strain (%)	R1 ($\text{\AA}$)	R2 ($\text{\AA}$)	E_b (eV)	M (μB)	E_g (eV)
-8	2.32	2.16	0.64	0.48	0.00
-6	2.33	2.17	-1.77	0.73	0.00
-4	2.33	2.19	-3.21	0.92	0.02
-2	2.34	2.29	-3.42	1.03	0.19
0	2.34	2.25	-3.41	1.12	0.08
2	2.34	2.23	-2.46	0.95	0.13
4	2.33	2.27	-0.73	1.05	0.00
6	2.29	2.29	1.33	1.34	0.00
8	2.25	3.52	1.15	0.39	0.00

stability of the SP structure, Fig. S1(a) depicted the variations in potential energy during a 50 ps molecular dynamics simulation at 298 K. The potential energy exhibited minimal fluctuation, indicating the dynamically stable presence of SP with no observable deformation under typical ambient conditions.

Fig. 2(a, c) showed the band structure of the BP and SP. The results testified that the pristine BP exhibited a semiconductor characteristic with a direct bandgap of 0.99 eV, which was in good agreement with previous theoretical studies [39,40]. The band structure of SP appeared some impurity bands and the conduction-band minimum (CBM) moved down to the Fermi energy, causing the band gap significantly decreased. Additionally, a deviation of the valence-band maximum (VBM) from the Γ -point indicated an indirect band gap in the SP. Although PBE functional was known for underestimating the bandgap of semiconductors, it was widely utilized in many studies due to its relatively accurate description of the changes in band structure under strain [41,42]. Meanwhile, in terms of sensing, the relative change in the band gap was deemed more significant than the absolute value [43,44]. Hence, considering the trade-off between computational cost and accuracy, GGA-PBE functional was used for the band structure calculation of the adsorbed systems.

The partial density of states (PDOS) in Fig. 2(d) revealed that the valence band near the Fermi energy was primarily originated from the Sc-d orbitals. It was also observed that distinct resonance peaks between P-p and Sc-d orbitals appeared near -0.71 eV, indicating the presence of

orbital hybridization. Thus, it could be concluded that Sc had a strong interaction with BP. Moreover, the asymmetric total density of state (TDOS) of SP also illustrated it belonged to a magnetic material with 1.12 μB magnetic moment [16]. This was attributed to the introduction of unfilled d orbitals in the Sc, leading to spontaneous spin polarization in the P sheet, thereby effectively regulated the properties.

To further understand the interaction between the Sc and BP, we calculated the charge density difference (CDD) of SP in Fig. 3. The electrons were accumulated around the P atoms. With the dipole correction in the charge transfer calculation, the Sc atom lost about 0.63 e, which were transferred to its three nearest-neighbour P atoms, resulting in the formation of strong Sc-P bonding. Additionally, the large bond overlap populations (Table 1) between Sc and three P atoms also indicated the great stability of the Sc-P covalent bonds.

3.2. Effect of biaxial strain on Sc-decorated black phosphorus

In this study, biaxial strain ranging from -8% to 8% was applied to SP. The geometric parameters and binding energies obtained from structural optimization were presented in Table 2, which reflected the effect of applying strain on the SP. As shown in Fig. S2, relatively stable geometric structures of SP were maintained within the strain range of -6% to 6% . The hexagonal structure of the BP layer under -8% and 8% collapsed due to disruption of the P-P bond structure within the layer. Particularly, SP exhibited the overlap population of -0.20 between P2 and P3 under the strain of -8% , reflecting a weak ionic interaction. Moreover, the binding energy of all structures fundamentally increased with the increasing of compression or stretching strains. The variation trend of binding energy was relatively gentle from 0 % to 4 % strain. Especially when the strain achieved more than 6 % or -8% , the binding energy of SP was positive which meant that it could not be prepared naturally, consistent with above the analysis of geometry. Therefore, SP exhibited a higher tolerance for compressive strain compared to tensile strain. These changes in structure likely had notable implications for the electronic properties of SP under strain.

To detailly investigate the electronic properties of SP under different biaxial strain, the band structure and DOS were calculated. Considering the stability of structures, we only researched the strains ranging from -6% to 6% . It was observed that the electronic properties of SP exhibited a strong dependence on strain. The band gap increased

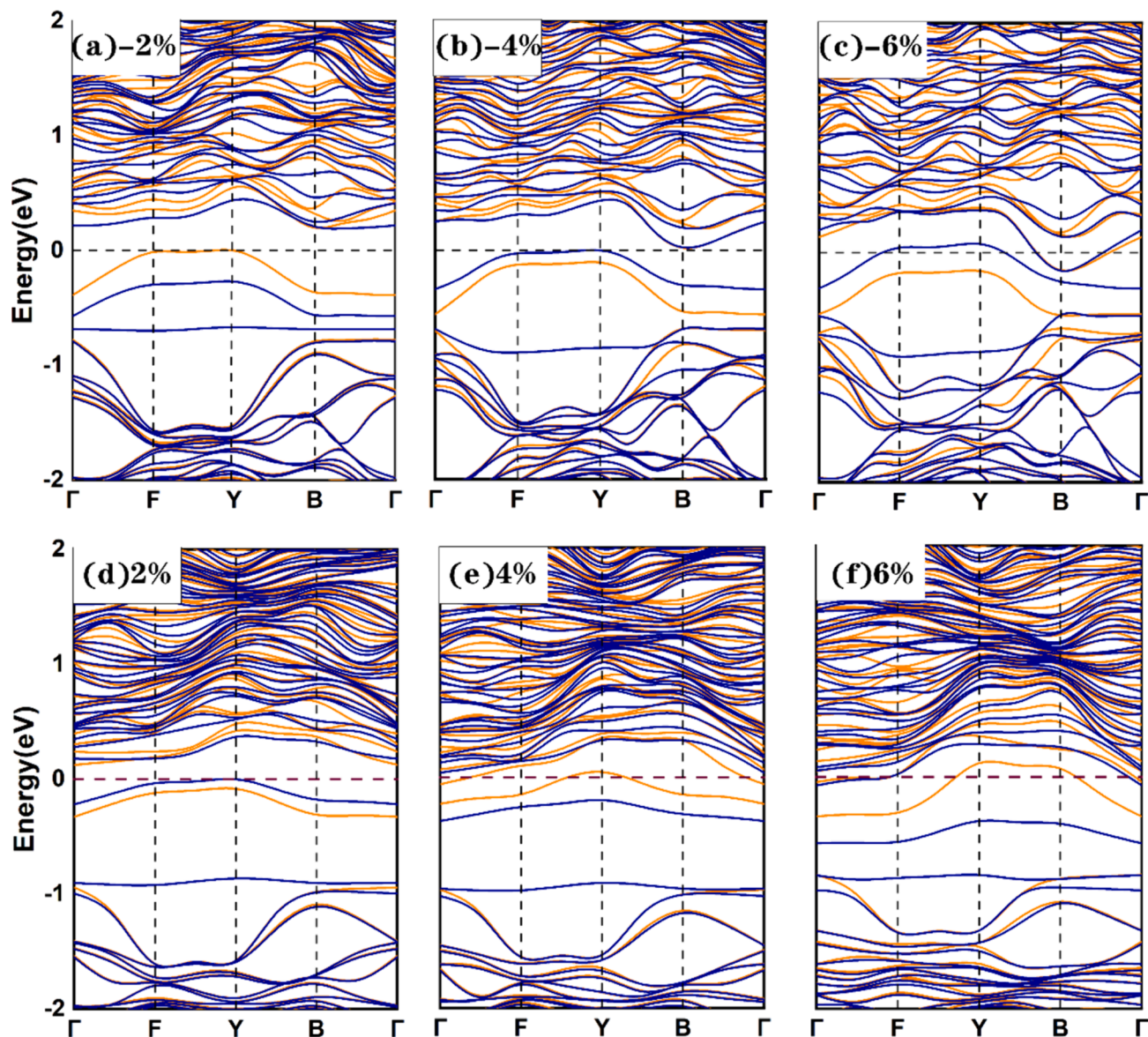


Fig. 4. Band structures of Sc decoration on monolayer BP under (a) -2%, (b) -4%, (c) -6%, (d) 2%, (e) 4%, (f) 6% biaxial strain.

initially and then decreased under both tensile and compressive strains, as shown in Fig. 4. At a strain of -4%, SP exhibited metallic behavior in one spin direction and semiconductor behavior in the other spin direction. This characteristic was known as spin half-metallic property [45]. With further increases of strain, the CBM and VBM crossed the Fermi energy and exhibited metallic behavior in both two spin directions. The asymmetric band structures for different spin directions indicated the occurrence of spin polarization in SP. Hence, the magnetic moment of SP under different strains were calculated, as listed in Table 2. The magnetic moment decreased with increasing compressive strain, or decreased first and then increased with increasing tensile strain.

Fig. 5 showed TDOS and PDOS of the SP systems, it was found that the contribution of Sc-d orbital to the energy levels significantly decreased under different strains. Also, the resonance peaks for the Sc-d and P-p orbitals became weaker and moved to lower energy with the larger strain, illustrating a low degree of orbital hybridization. The results demonstrated a non-monotonic behavior of the band gap values and basically decrease of the interaction between Sc and P as the tensile and compressive strains increasing. This suggested that the electronic properties of SP exhibited a strong dependence on strain and could be tuned effectively by strain.

3.3. Gas sensing of Sc-decorated black phosphorus

NO gas was a common air pollutant in automobile exhaust and industrial activities. It was prone to explode with combustibles and poses health risks [46]. Hence, Gas sensors were vital for protection systems in industrial and household settings [10]. Based on this background, we conducted simulations to analyze NO molecule adsorption on BP and Sc-decorated BP.

For the NO molecule adsorption system, different adsorption sites may affect the stability and electronic properties. Firstly, H, T and B adsorption sites were considered on BP, as shown in Fig. S3(a-c). The NO molecule was finally stabilized at the T site of BP substrate with an adsorption energy of -0.21 eV (Table S2), which was closed to previous reports [47,48]. Meanwhile, three adsorption configurations for SP were showed in Fig. S3(d, e), with the N-O bond either parallel or vertical to the BP plane, and with N or O pointing towards Sc. The NO molecule exhibited a preference for the proximity of the N atom to the metal atom owing to the higher levels of adsorption energy.

The optimized structures of adsorption NO molecule on BP (BP-NO) and SP (SP-NO) were shown in Fig. S3 and Fig. 1(c) respectively. The adsorption of NO molecule on SP showed a noticeable reduction in

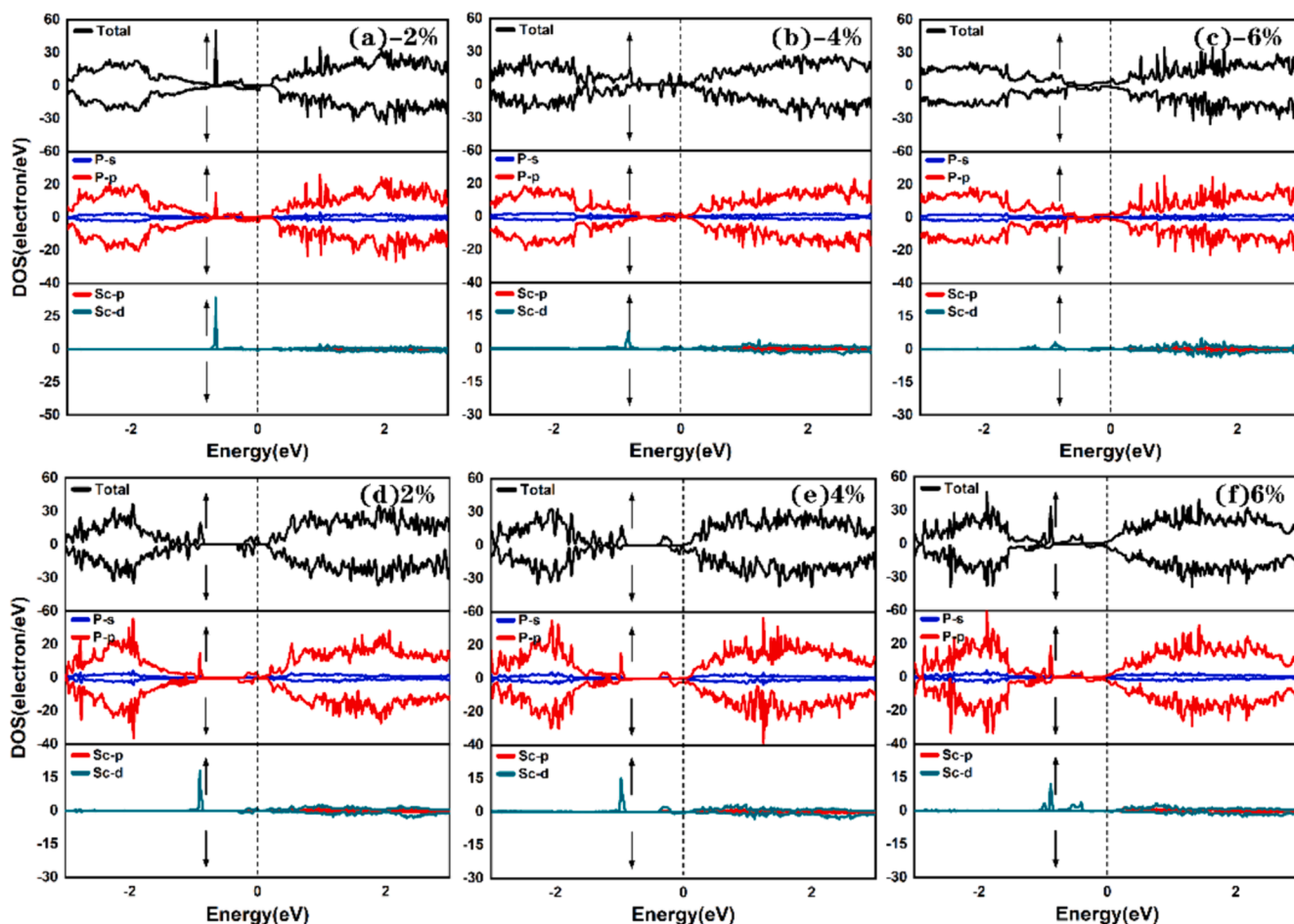


Fig. 5. TDOS and PDOS of SP under (a) -2% , (b) -4% , (c) -6% , (d) 2% , (e) 4% , (f) 6% biaxial strain. The Fermi energy was set as zero.

adsorption height and adsorption energy with -2.92 eV compared to BP. The covalent radii of Sc and N atoms are 1.48 Å and 0.7 Å, respectively, with a total sum of 2.18 Å [49]. The adsorption distance of the molecule was 2.35 Å in BP-NO and 1.92 Å in SP-NO away from the substrate surface, respectively exceeding and falling short of 2.18 Å. In general, physisorption referred to the condition where the absolute adsorption energy was less than 0.4 eV, and the closest atomic distance between the adsorbed molecule and the substrate was greater than the sum of the covalent radii of the corresponding atoms [50]. In addition, the RDF peak at 1 Å to 3.5 Å indicated short bonds for chemisorption, while peaks over 3.5 Å were linked to physical interactions [51]. Fig. S1 showed the peaks of RDF were at 7.03 Å and 7.23 Å for BP-NO, and a peak at 1.90 Å for SP-NO. A higher overlap population value indicated a higher level of covalence. A small overlap population of the P1-N bond in BP-NO was 0.05 , indicating a weak interaction between BP and NO molecules. However, the N-Sc overlap population in SP-NO significantly increased to 0.53 , as shown in Table 1. To sum up, we could speculate NO molecule adsorption on substrate turning from weak physics adsorption into the strong chemical adsorption.

The adsorption mechanisms were analyzed from two aspects of charge transfer and orbital hybridization, utilizing methods such as Mulliken charge transfer, charge density difference, and DOS. Mulliken charge distribution analysis was employed to discover that SP transfers more electrons (0.47 e) to NO molecule compared to BP (0.30 e), which further illustrated the strong interaction between SP and NO. The CDD revealed that significant charge accumulation and dissipation occurred between the NO molecule and SP, as shown in Fig. 3(c). Meanwhile, the PDOS in Fig. 6 revealed weaker resonance peaks between the P-p orbital and N-p orbitals near the Fermi energy in BP, while clear peaks were

observed between Sc-d and N-p orbitals in SP, indicating notable orbital hybridization and significant charge transfer between the Sc atom and the N atom. Furthermore, the potential energy from the molecular dynamic simulation (Fig. S1(b)) suggested it remained stable and did not show significant distortion under ambient conditions. As a result, Sc-decorated black phosphorus could significantly improve the adsorption capability to NO gas molecule with excellent structural stability.

The band structure was depicted in Fig. 6. In the case of BP-NO, it could be found a slight reduction in the band gap and a shift to an indirect band gap after NO adsorption (Fig. 6(a)). The calculated magnetic moment of 1.00 μ_B was consistent with the findings in reference [52]. For SP-NO, the band structure depicted in Fig. 6(c) demonstrated a reduction in impurity states and a movement of the Fermi energy towards the valence band, leading to a significant increase in the band gap. Meanwhile, the symmetric band structure confirmed the disappearance of magnetic properties and the closure of spin polarization.

The gas-sensing properties of BP and SP towards NO gas were further explored based on the following principle, with a focus on their sensitivity, work function and recovery time. The conductivity of SP monolayer would change due to the band gap variations described above. And the sensor used field-effect transistors (FETs) with integrated material to detect gas molecules. When the gas molecules interacted with the substrate surface, it caused a change in the conductivity of the FET by altering the density of charge carriers. One important indicator that determined the quality of gas sensors was sensitivity, which was calculated as follows [18,53]:

$$S = 100 \times \left(\frac{|\sigma_0 - \sigma|}{\sigma_0} \right) \quad (4)$$

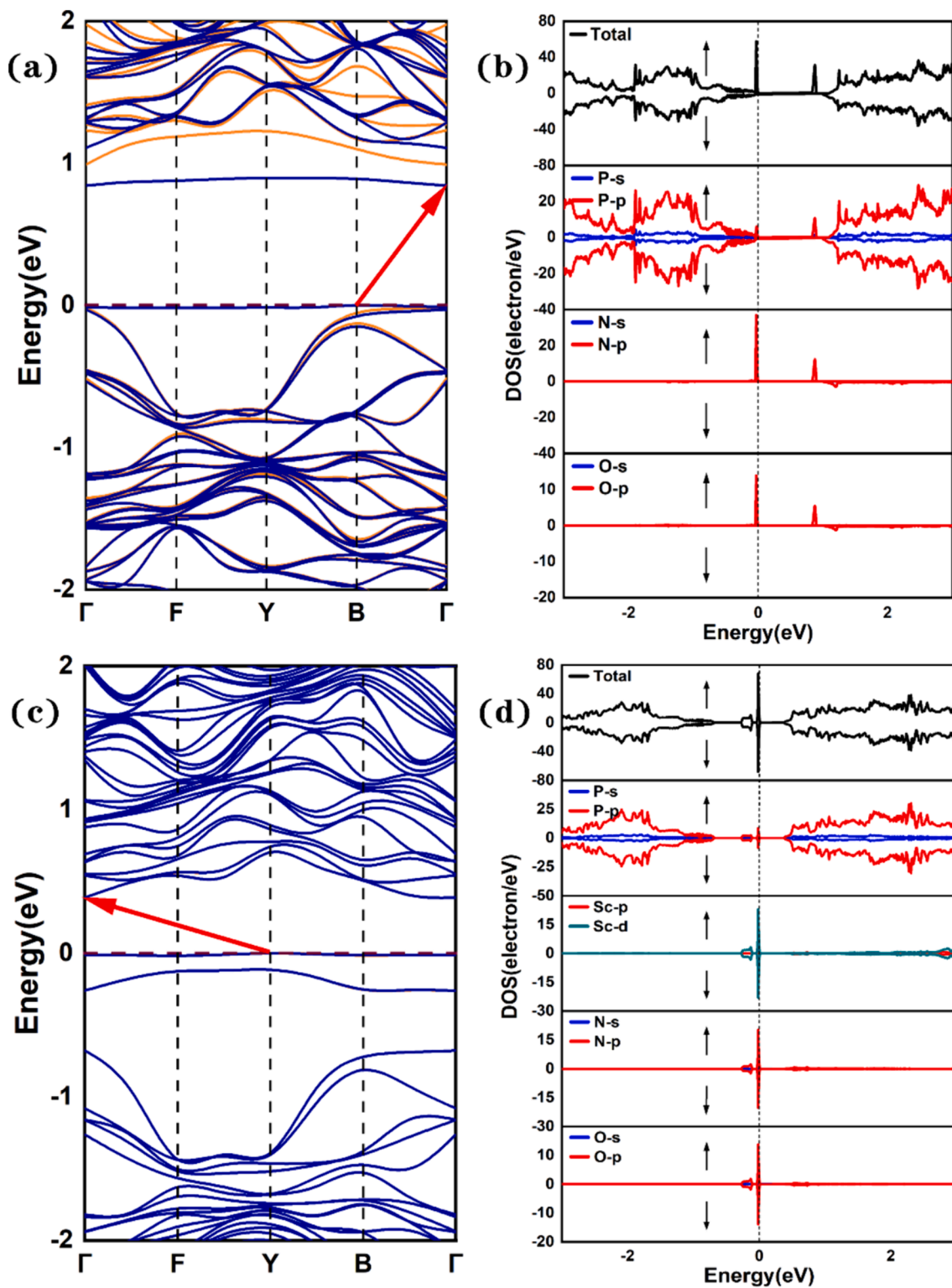


Fig. 6. (a) The band structure and (b) spin polarized DOS of BP-NO. (c) The band structure and (d) spin polarized DOS of SP-NO. The dash line in the DOS illustrated the Fermi energy.

where S represented gas sensitivity, σ denoted the conductivity of the sensor during operation, and σ_0 represented the zero-bias conductivity of the sensor before adsorption. The conductivity (σ) was related to the band gap of the system [54]:

$$\sigma \propto T^3 \exp\left(\frac{-E_g}{2k_B T}\right) \quad (5)$$

where T represented the operating temperature (K), k_B denoted the Boltzmann constant ($8.62 \times 10^{-5} \text{ eV} \cdot \text{K}^{-1}$).

The adsorption of NO molecule significantly changed the band gap of SP, resulting in a large change in conductivity and an increasing gas sensitivity towards NO molecule. As shown in Table S2, the gas sensitivity (298 K) of BP towards NO molecule was relatively low (25.2), while the sensitivity of SP (82.1) was 3.3 times that of BP, indicating a

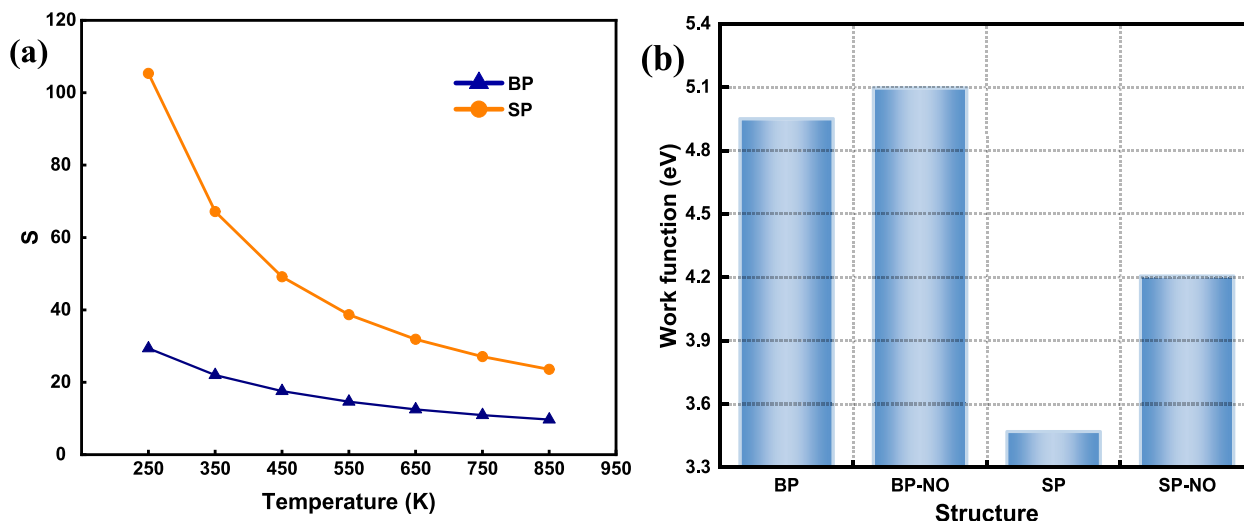


Fig. 7. (a) Effect of temperature on the sensitivity (S) of BP/SP substrates to NO gas and (b) work function for NO gas molecule adsorption BP/SP surface.

Table 3

In the case of BP adsorbing NO and ranging from −8% to 8% strains of SP-NO, the height between the nearest atom towards the NO molecule of substrate and the N atom (L_h), the bond lengths of the P-P bonds in the BP layer in both the vertical and horizontal directions (R1, R2), the gas adsorption energy (E_{ads}).

System	Strain (%)	L_h (Å)	R1 (Å)	R2 (Å)	E_{ads} (eV)
BP-NO		2.34	2.27	2.24	−0.21
SP-NO	−8	1.37	2.30	2.16	1.55
	−6	1.31	2.30	2.17	−1.03
	−4	1.28	2.30	2.19	−2.60
	−2	1.26	2.30	2.21	−3.21
	0	1.33	2.23	2.29	−2.92
	2	1.27	2.28	2.26	−1.97
	4	1.30	2.28	2.29	−0.37
	6	1.37	2.30	2.30	1.68
	8	1.49	2.26	2.33	4.05

significant improvement in the sensitivity towards NO gas. Additionally, the magnetic moment in SP-NO disappeared to attain a non-spin state, serving as a criterion for distinguishing NO gas. Since temperature could alter conductivity [55], the gas sensitivity of BP and SP under different temperature was investigated, as shown in Fig. 7(a). At 850 K, the gas sensitivity of BP and SP respectively decreased by 66.9 % and 77.6 % while the gas sensitivity of SP remained 2.4 times higher than that of BP. Hence, SP still exhibited good gas sensitive performance at high temperature.

The adsorption of nitric oxide may also cause a modification in the work function of the chemiresistive nanodevice, which could evaluate its potential as field-effect transistor (FET) type gas sensor. The work function was commonly represented by the following expression [11,18]:

$$\Phi = E_{vac} - E_F \quad (6)$$

where E_{vac} denoted the vacuum energy, which was the electrostatic potential in the middle of vacuum, and E_F signified the Fermi energy. The measured values were depicted in Fig. 7(b). The calculated work function for the pristine BP layer was found to be 4.95 eV, which was reasonably close to the theoretical value of 5.03 eV [32]. The significant reduction of work function after Sc decoration suggested easier electron release to vacuum, facilitating their participation in surface reactions. In terms of adsorption of NO gas molecule, the SP sensor demonstrated a more pronounced enhanced in work function by 21.21 %, while BP sensor increased by 2.95 %. It was found that the work function sensitivity of the Sc-decorated BP increased by a factor of 7, suggesting the

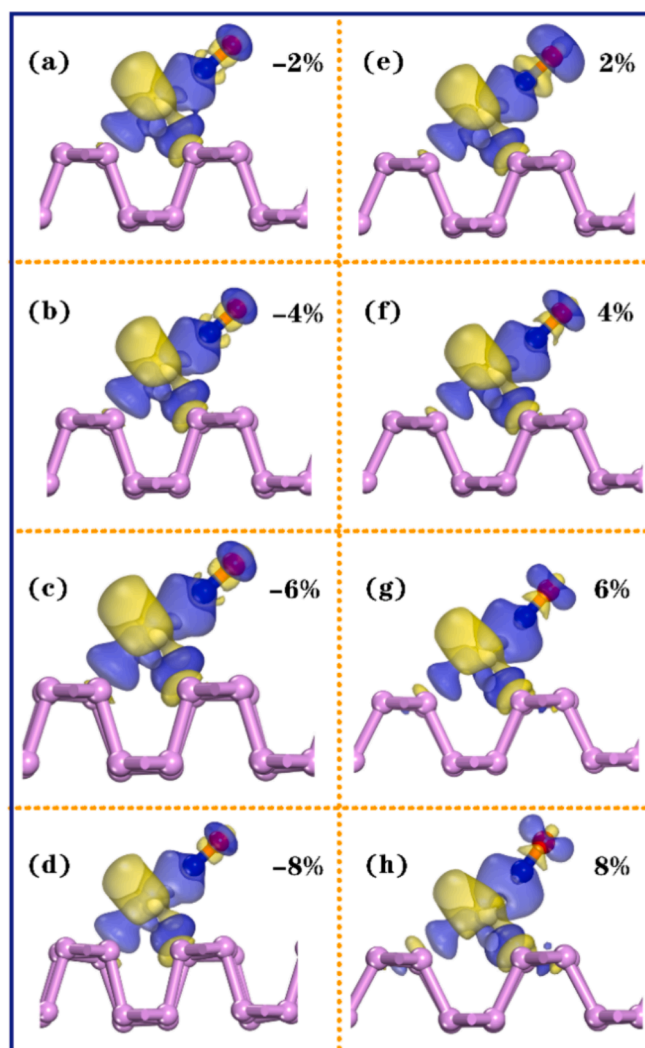


Fig. 8. The CDD of SP-NO under the biaxial strain of (a) −2%, (b) −4%, (c) −6%, (d) −8%, (e) 2%, (f) 4%, (g) 6%, (h) 8%.

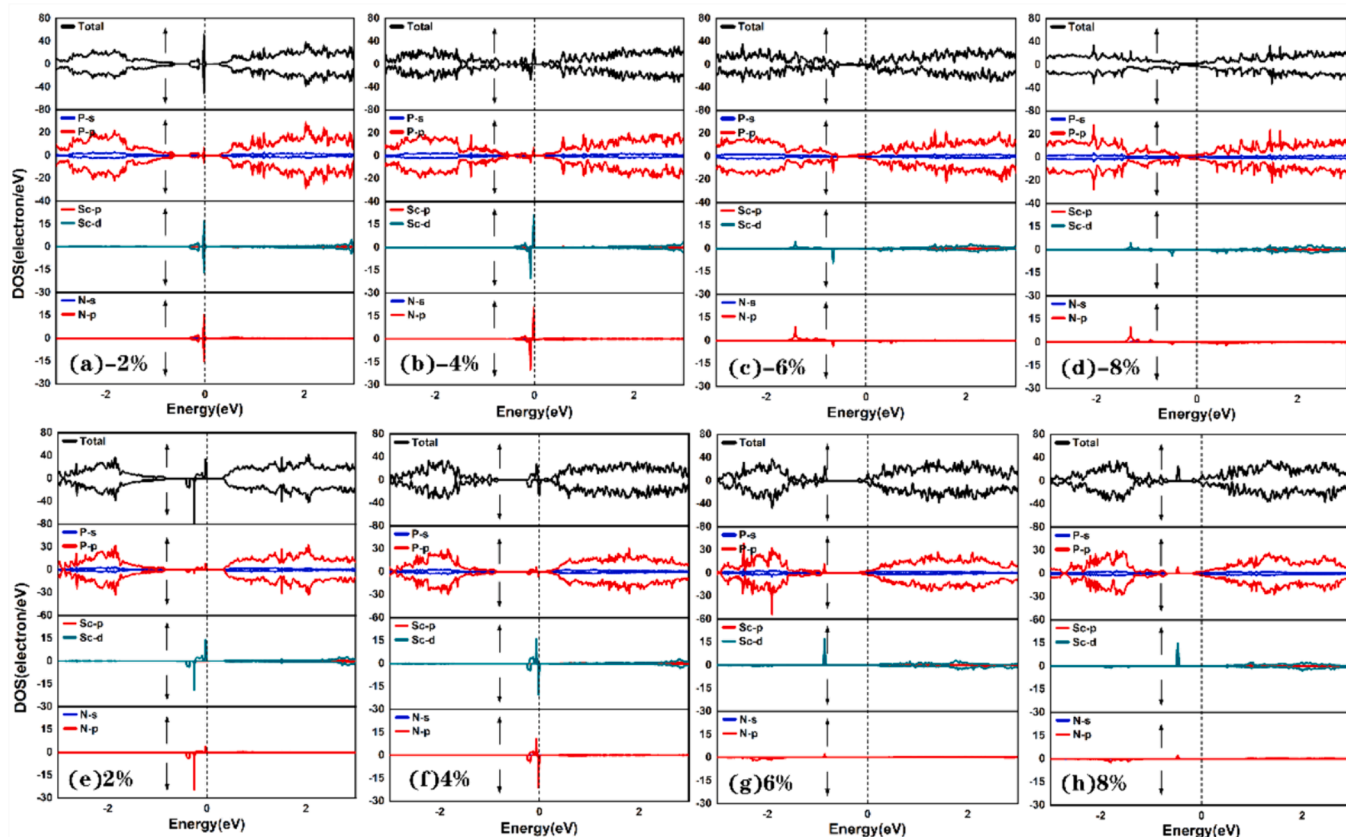


Fig. 9. The TDOS and PDOS of SP-NO under the biaxial strain of (a) -2%, (b) -4%, (c) -6%, (d) -8%, (e) 2%, (f) 4%, (g) 6%, (h) 8%.

potential for SP to be used as a work function-sensitive type gas sensor.

The reusability of gas sensors was one of the most critical parameters to evaluate their performance. The recovery time was calculated using the traditional transition state theory (TST) with the adsorption energy [56]:

$$\tau = \nu_0^{-1} \exp\left(\frac{-E_{ads}}{k_B T}\right) \quad (7)$$

where ν_0 represented the attempt frequency (10^{12}s^{-1}), and E_{ads} represented the adsorption energy (kcal/mol), as derived from the previous Eq. (3). The pristine BP was unsuitable for gas detection under 298 K due to its recovery time (3.56 ns), which was too brief for stable operation. Particularly, SP had a significant adsorption capacity due to its large adsorption energy with NO molecule, resulting in a prolonged recovery time. Therefore, SP could be used as a highly sensitive functional and disposable sensor for NO gas molecule and serving as a powerful capturer of NO gas molecule with the adsorption energy of -2.92 eV.

3.4. Adsorption of NO gas on Sc-decorated black phosphorus under biaxial strain

For the purpose of substrate recycle, it was important to explore efficient and environmental-friendly ways for desorbing NO molecule from substrate. Some previous works have shown that applying strain or electric field could change the interaction strength between gas molecule and substrates [43,57]. In this section, NO molecule adsorption on SP was analyzed under the biaxial strain in the range from -8% to 8% to assess the potential for switchable adsorption and desorption. The geometric parameters and adsorption energy of SP-NO under different strains were shown in Table 3. It was found that the overall geometric structure of the SP substrate remained nearly unchanged when strain was applied to SP-NO. At the same time, the adsorption height of NO

molecule on SP generally exhibited an increasing trend with the increase of strain, resulting in the weakness of adsorption energy. In particular, the positive adsorption energy under the strain of -8% and 6% indicated the interaction between NO molecule and SP became electrostatic repulsion, and NO molecule could be released from the substrate without any barrier. Furthermore, the geometric parameters of the BP layer in the SP-NO system recovered back to pre-strain state under 8% strain, while in the SP system, severe subsidence of Sc caused significant deformation of the BP layer. It revealed that structural stability of SP significantly enhanced after NO molecule adsorption, which made it be reused after the NO molecule release. Thus, the desorption of NO molecule could be achieved by applying the biaxial strain. In this way, the purpose of substrate recycling was easily realized.

To further research the adsorption and desorption mechanism of strain controlling the desorption of NO molecule on SP, the CDD and DOS of SP-NO under different strains were carried out. The gas interaction with SP-NO could be understood by the CDD, which helped to predict the charge redistribution behavior between the substrate and gas. As shown in Fig. 8(a), it showed an increase in the overlap region around N atom when applied strain of -2%, indicating enhanced electronic interactions between atoms, and improved structural stability of SP-NO. The overlap region of electron around N atom reduced in varying degrees with strain increasing, leading to a weak interaction, confirming the previous trend in adsorption ability reduction. The adsorption energy became positive under the strain of -8% and 6% since the electron redistribution would occur in the adsorption systems through the modulation of the applied biaxial strain. It led to the strong interaction between NO molecule and SP turning to weak interaction.

Fig. 9. presented the TDOS and PDOS of NO gas molecule adsorption on SP under the different biaxial strains for critical analysis of adsorption and desorption. The asymmetric TDOS indicated that the magnetism has been activated under the strain. The PDOS revealed that both

Sc-d and N-p orbitals spin split and the distance between spin-up and spin-down peaks increased under strain. At the same time, the resonance peaks between N-p and Sc-d orbitals showed a decline and moved towards the lower energy, illustrating the interaction between NO molecule and SP became weaker. In particular, the resonance peak of N-p orbital basically disappeared under the strain of 6 % and 8 %, which further explained the desorption of NO molecule. It could be concluded that the modulation of biaxial strain could effectively govern the capture and release of NO molecule. Meanwhile, this technique facilitated the attainment of the objective to recycle the substrate materials, promoting their sustainable utilization.

4. Conclusion

In summary, the electronic properties of Sc-decorated black phosphorus and its sensing capability towards NO gas molecule were comprehensively investigated by first-principles methods. Particularly, the effect of biaxial strain was further researched. The appearance of impurity states turned Sc-decorated black phosphorus into an indirect band gap semiconductor and significant decrease of band gap led to a high electrical conductivity. The application of biaxial strain resulted in a controllable magnetic moment while induced a transition from a semiconductor to metallic properties. Interestingly, Due to orbital hybridization and extensive charge transfer of 0.47 e, Sc-decorated black phosphorus exhibited a high NO molecule adsorption energy of -2.92 eV, indicating significant adsorption capability. Simultaneously, the band gap and work-function changes upon NO molecule adsorption on Sc-decorated increased by 3.3 and 7.2 times, respectively, compared to black phosphorus, showcasing its heightened sensitivity. Furthermore, the application of biaxial strain induced the positive adsorption energy, bringing about desorption of NO molecule. Simultaneously, the geometric structure reverted to pre-strain state indicating that structural stability of Sc-decorated black phosphorus remarkably boosted after NO molecule adsorption, which made it be reused after the NO molecule release. In conclusion, we demonstrated the modulating effect of strain engineering on the electronic properties and gas sensing of the Sc-decorated black phosphorus, providing a novel theoretical guidance for the practical design of reversibility gas sensor.

CRedit authorship contribution statement

Houren Chen: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Longying Wen:** Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Zilin Chen:** Writing – review & editing, Validation, Software, Investigation, Conceptualization. **Hao Jiang:** Investigation, Software, Validation. **Huadong Zeng:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cplett.2024.141303>.

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